INTELLIGENT ELEVATOR CIRCUIT

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DESCRIPTION

- We design an elevator has 4 floors using digital circuits to control the movement of it , we can call any floor ,but must proceed through intermediate floor before reaching the destination .
- the logic gates in the elevator system are designed to perform specific functions, such as comparing input signals , generating output signals based on certain conditions ,and controlling the movement of the elevator motor (up , down , still in the same floor) .
- By using logic gates , the elevator system can be programmed to perform complex operations with high accuracy and reliability.

- We input the current floor (using D_flip flop) and the destination floor using input button, so it will show if we go up , down or still in the same floor.
- the elevator system typically consists of light sensors that detect the position of the elevator car ,if the red led (up) illuminates the sensor follow the light and go from the current floor to the our destination ,and we show the change of the floor on the hexa_segment.

--OUR TASK IS TO DESIGN A SEQUENTIAL CIRCUIT THAT WOULD IMPLEMENT THE STATE MACHINE FOR THE ELEVATOR CONTROLLER
*START BY SAYING THE STATE AND INPUT OF THIS DEVICE.

- State of the device

  -G , 1 , 2 , 3.

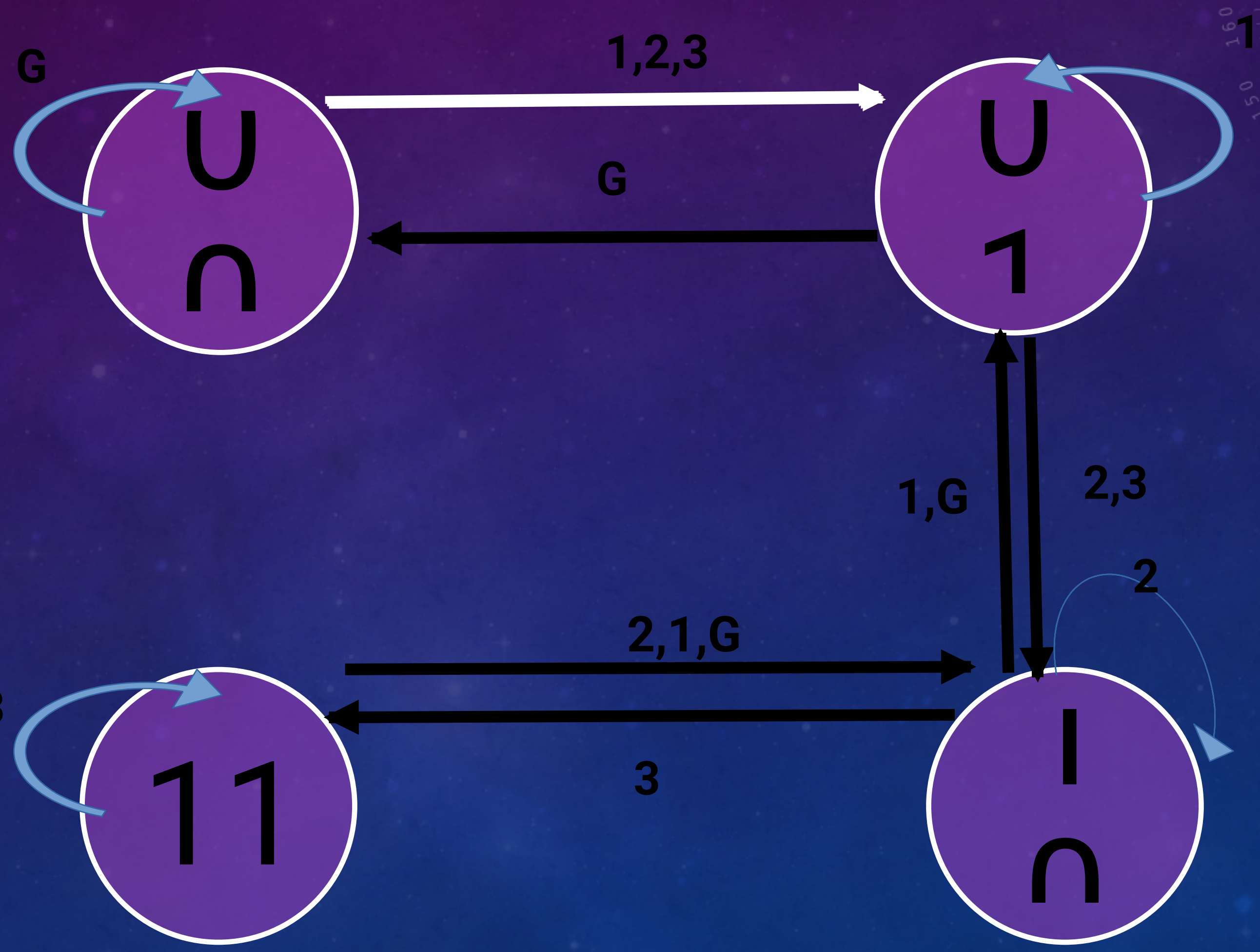
  -00 , 01 , 10 , 11

- Input (go to the …….)

  -G , 1 , 2 , 3.

  -00 , 01 , 10 , 11

THE STATE DIAGRAM FROM (00 ⟶ 11)



EXPLAIN STATE DIGRAM

- If I am in the ground (state) and need to go the third floor (input) the elevator must go from ground to first floor and from first to second till I reach my destination.

Truth Table of State and Next State

Q1Q0	F1f0	Q1^+ Q0^+
00	00	00
00	01	01
00	10	01
00	11	01
01	00	00
01	01	01
01	10	10
01	11	10
10	00	01
10	01	01
10	10	10
10	11	11
11	00	10
11	01	10
11	10	10
11	11	11

THE Q 1 plus and Q0 plus k-map

F1 F0

Q1Q0

0	0	0	0
0	0	1	1
1	1	1	1
0	0	1	1

D1 =Q1+=Q1Q0+Q1F0+Q0F1

F1 F0

Q1Q0

0	1	1	1
0	1	0	0
0	0	1	0
1	1	1	0

D2=Q0+=Q0(bar)F0+ F0(Q1 XNORF1) +Q0(BAR)*(Q1 XOR F1)

THE USED GATES

1. 6 AND gates
2. 1 NOT gate
3. 2 OR gate
4. 1 XNOR
5. 1 XOR
6. 2 D_flip flop
7. 12 Comparator
8. 8 Hexa_Segment

Explaining segments

Here in our project to Detect Each floor we suggest an a sensor to follow the first led then
Enter the detected led circuit .

– if the stop led (the result of comparing the present state and the next state)
on then after combaring the present state and the Required state
The result is equal and the secound led of stop on .

– if if the up led (the result of comparing the present state and the next state)
on then after combaring the present state and the Required state
The result is Greater than and the secound led of up on
And then enter the up circuit .

Explaining the Segments

** in the up circuit
We compare the required floor and the present floor and follow the led
 * for Ex:
If we wand the 3rd floor and we are in G floor then the first up
 led is on (the result of comparing the next state (01)and the present state (00)
Then we comparing the next state(01) and the required floor (11)and the Result is up
led is on
Then we enter in the up circuit and follow the led on Comparing and minus 1 ..

Explaining segments

– if if the Down led (the result of comparing the present state and the next state) on then after combaring the present state and the Required state
The result is Greater than and the secound led of up on
And then enter the up circuit

** in the Down circuit
We compare the required floor and the present floor and follow the led
* for Ex:
If we wand the G floor and we are in 3rd floor then the first Down led is on (the result of comparing the next state (10)and the present state (11)
Then we comparing the next state (10)and the required floor (00) and the Result is Down led is on
Then we enter in the Down circuit and follow the led on and Comparing and minus 1 .

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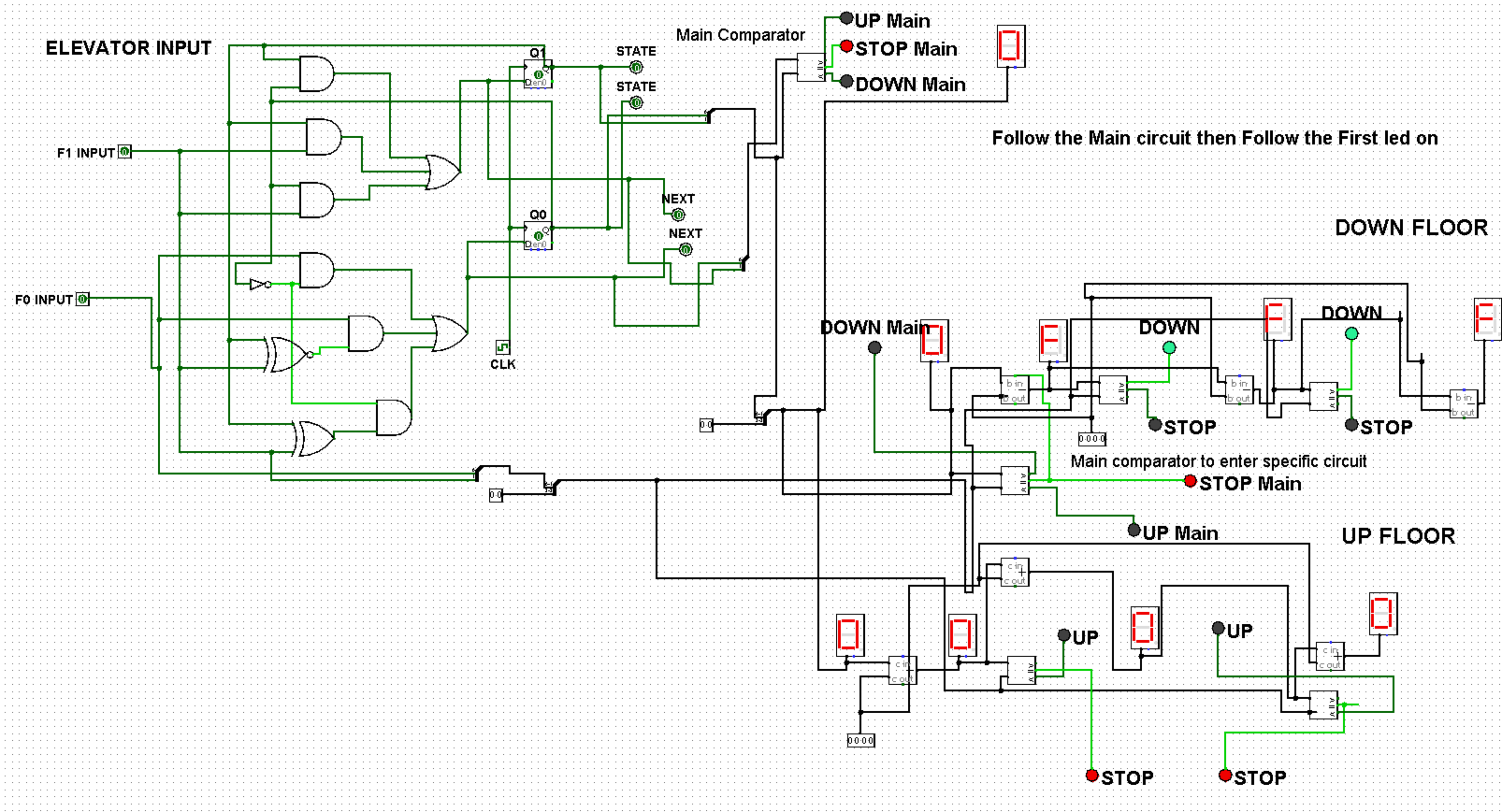

THANK YOU